

Safety Data Sheet

Ethanol (Ethyl Alcohol), ≥ 95%

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Section 1 — Identification

Product name: Ethanol (ethyl alcohol). Synonyms: rectified spirit, EtOH. CAS No.: 64-17-5. EC No.: 200-578-6. Recommended use: solvent; fuel/bioethanol intermediate. Emergency telephone (site): ext. 2222.

Section 2 — Hazards Identification

GHS classification: Flammable liquid, Category 2 (H225 — Highly flammable liquid and vapour); Eye irritation, Category 2A (H319); Specific target organ toxicity, single exposure, Cat 3 (H336 — may cause drowsiness or dizziness). Signal word: DANGER. Pictograms: flame, exclamation mark.

Precautionary: Keep away from heat, sparks, open flames, hot surfaces — no smoking (P210). Ground/bond container and receiving equipment (P240). Use explosion-proof electrical equipment (P241). Wear eye protection (P280).

Section 4 — First-aid Measures

Route	Action
Eyes	Rinse cautiously with water for several minutes; remove contact lenses if present and easy. Continue rinsing. If irritation persists, get medical advice.
Skin	Remove contaminated clothing; wash with soap and water.
Inhalation	Move to fresh air; keep at rest. If breathing is difficult, get medical attention.
Ingestion	Rinse mouth. Do NOT induce vomiting. Seek medical attention.

Section 5 — Fire-fighting Measures

Suitable extinguishing media: alcohol-resistant foam, dry chemical, carbon dioxide (CO₂). Unsuitable: DO NOT use a solid/straight water jet — it may spread the fire; ethanol is water-miscible and burns with a pale, sometimes barely visible flame. Cool exposed containers with water spray. Vapours are heavier than air and can travel to an ignition source and flash back.

Section 7 — Handling & Storage

Handle in well-ventilated areas away from ignition sources. Bond and earth all equipment to prevent static discharge. Store in a cool, well-ventilated place in tightly closed, approved containers, away from oxidisers. Keep container closed when not in use.

Section 8 — Exposure Controls / PPE

Basis	Value
Occupational exposure limit (8-h TWA, typical)	1000 ppm (1920 mg/m ³)
Eye/face	Chemical splash goggles
Hands	Chemical-resistant gloves (e.g., nitrile for splash)

Basis	Value
Body / respiratory	FR coveralls; use respiratory protection if vapours exceed the OEL

Section 9 — Physical & Chemical Properties

Property	Value	Property	Value
Appearance	Colourless liquid	Odour	Characteristic, alcoholic
Flash point	~13 °C (closed cup)	Boiling point	~78 °C
Auto-ignition	~363 °C	Flammable limits	3.3 – 19 % v/v in air
Relative density	0.79 (water=1)	Vapour density	1.6 (air=1)

Section 10 — Stability & Reactivity

Stable under normal conditions. Incompatible with strong oxidising agents, acids, and alkali metals. Hazardous reactions can occur with strong oxidisers. Avoid heat, flames, sparks.